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REVIEW ARTICLE

Sleep Monitoring in Athletes: Motivation, Methods, Miscalculations and Why it Matters

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Abstract

It is increasingly recognised that many athletes have poor sleep quantity and/or quality despite the advances in knowledge regarding the importance of sleep for an athletic population. The majority of research in sleep assessment and treatment (within the general population) focuses on the medical disorder insomnia, and therefore may not be specifically relevant for athletes. Further, there are currently no guidelines for the standardisation of assessment, intervention, feedback and behaviour change strategies in athletes. This review outlines potential reasons for sleep disturbances in athletes, advantages and disadvantages of a range of methods to assess sleep (polysomnography, activity monitoring, consumer sleep technology, sleep diaries and questionnaires), considerations for the provision of feedback, a description of potential interventions and behaviour change challenges and strategies. The objective of this review is to provide practitioners with the latest scientific evidence in an area rapidly progressing in awareness, consumerism and athlete engagement.

Key Points

There are currently no guidelines or recommendations for best practice regarding monitoring of sleep in athletes.

There are numerous advantages and disadvantages of available methods to measure sleep in athletes. It is critical that there is an understanding of the limitations of methods, including activity monitoring, consumer-based technologies and smartphone applications to ensure appropriate data interpretation and feedback.

Cognitive behaviour therapy for insomnia is recommended as the initial treatment for insomnia and future investigations in behavioural science to optimise changing sleep behaviours in athletes are warranted.

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1 Introduction

Sleep is becoming widely recognised as essential for optimal heath, wellbeing and athlete performance. Alongside this increased recognition is a concomitant increase in the desire to monitor athletes' sleep and address presenting sleep concerns or disturbances. As a consequence, there is now a multitude of consumer-based technologies (commercial sleep technology [CST]) and smartphone applications available in addition to the gold standard of sleep assessment (polysomnography), research-grade activity monitoring and sleep diaries and questionnaires.

Research conducted examining sleep quality and quantity in athletes has typically utilised activity monitoring with most studies reporting athletes demonstrate insufficient sleep (below the recommended 8 h per night) and many sleep poorer than the general population [1]. Given the reported effect of sleep deprivation on performance, mood, cognitive function, memory, learning, metabolism, illness and injury [2], it is of no surprise that assessing sleep in athletes has become increasingly popular.

While there are a number of international consensus statements, position stands and guidelines for managing and treating insomnia, obstructive sleep apnoea (OSA) and other medical sleep disorders [3–5], there is little information on how to best measure, analyse, provide feedback and intervene in athletes who do not have a clinical sleep disorder and

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where monitoring is utilised as a tool to optimise athlete performance. Therefore, the purpose of this article is to provide sport science and sport medicine practitioners with current best practice recommendations for monitoring sleep, including advantages and disadvantages of available monitoring tools, the provision of feedback and potential interventions.

2 Presenting Reasons for Insufficient Sleep in Athletes

In order to gain a comprehensive understanding of suitable monitoring tools and potential effective interventions, it is worth noting that there are a variety of non-medical explanations for poor sleep in athletes. Some of these aspects of an elite athlete's lifestyle are more difficult to manage, while others may be addressed through appropriate education and behaviour change strategies.

Stress and anxiety have been comprehensively reported in the scientific literature to negatively affect sleep in the general population [6] and competition stress has great potential to interfere with sleep in athletes. Juliff et al. [7] investigated a sample of 283 elite Australian athletes, with 64.0% of athletes indicating they slept worse than usual on at least one occasion in the nights prior to an important competition over the past 12 months. The main sleep concern identified by athletes was problems falling asleep (82.1%) with the main reasons responsible indicated as 'thoughts about the competition' (83.5%) and 'nervousness' (43.8%). Anecdotally, stress and anxiety from other sources such as team selection, sponsorship arrangements, work or school stress and relationship issues may also result in poor sleep.

Training and competition times may also influence sleep, with early morning training times resulting in reduced sleep in elite swimmers [8]. Late night games in professional team sport athletes may also result in shortened sleep. Sargent and Roach [9] compared the sleep of Australian Rules Football players following a day game and a game played at night. After the evening game, sleep onset was later, time in bed was shorter and total sleep obtained was less than after the day game. Similar impaired sleep was also reported by Lalor et al. [10] and Dunican et al. [11]. Of particular concern, in the latter study in a Super Rugby team, four of the 20 players did not attempt sleep on the night after a game [11]. This information may not have been known by the coaching and support staff had the activity monitoring not occurred.

The intensity of training may also influence sleep; while one would anticipate that sleep would improve (due to an increased need) during intensified training, evidence suggests that this does not occur. In a 3-week simulation of a cycling Grand Tour, Lastella et al. [12] reported that the amount and quality of sleep as assessed by activity monitoring declined. Similar observations of reduced sleep following intensified training were reported by Killer et al. [13] and Hausswirth et al. [14].

Other factors such as caffeine consumption, muscle soreness, injury, jetlag, travel (sleeping in foreign environments) and social media/video games are anecdotally reported to have a profound influence on an athlete's sleep if not managed effectively.

3 Monitoring Tools

3.1 Polysomnography

Polysomnography (PSG) is considered by many to be the gold standard for measuring sleep and typically includes an assessment of eye movement, brain activity, heart rate, muscle activity, oxygen saturation, breathing rate and body movement [15]. PSG allows for the determination of sleep stages (rapid eye movement [REM] and non-rapid eye movement [NREM]). PSG is typically only utilised to assess specific sleep disorders as it is an expensive, obtrusive and complex technique, requiring expertise in setting up the individual to be monitored and manually scoring the sleep data. Further, PSG typically occurs in a laboratory and therefore is impractical for longitudinal monitoring. According to the Australasian Sleep Society, PSG is not recommended for the assessment of insomnia, restless leg syndrome or uncomplicated parasomnias, but may be useful if there is a suspicion of OSA or overlapping sleep disorders [5]. Finally, PSG may be useful in a research context where understanding the effects of an intervention on sleep staging may be of interest.

3.2 Activity Monitoring/Actigraphy

Activity monitors are small, wearable devices which record movement as a function of time [16], with most utilising a 3-axis accelerometer to determine sleep/wake based on a specific algorithm [17]. Research-grade activity monitors (e.g. Philips Respironics ActiWatch 2) have been validated against PSG and the algorithms utilised to quantify sleep and wake are reported. Importantly, activity monitoring has been validated in an athlete population [18, 19]. The Philips ActiCal device, when set to a medium threshold for detection of sleep/wake, demonstrated a mean bias (with PSG) of 8.5 min in total sleep time, 1.8% for sleep efficiency and −4.1 min for wake after sleep onset [19]. Good agreement (81–90%) and sensitivity (81–92%) between activity monitors (Philips ActiCal) and PSG have also been observed in athletes [18]. Activity monitors underestimated sleep duration (18–90 min) and overestimated wake duration (4–77 min) depending on the threshold applied. In this study, thresholds with high sensitivity to sleep (> 80 activity counts = wake)

resulted in the best combination of agreement, sensitivity and specificity when compared with PSG [18].

Actigraphy allows for longitudinal monitoring (beneficial for assessment of sleep/wake times across numerous days) and, due to a low subject burden in comparison with PSG and the elimination of recall and reporting biases, they are popular devices in both research and practice [16].

The scoring algorithms for sleep and wake have been shown to perform well in healthy sleepers, with deficiencies lying mostly in their ability to detect wake, rather than sleep. As such, they perform more poorly in those individuals with fragmented sleep who have higher amounts of wake during sleep [16]. Further, research quality devices can be expensive (approximately US$1000 per unit, plus software) and some expertise is required to ensure accuracy of the data provided and to consolidate these data into a usable sleep report.

3.3 Commercial Sleep Technology (CST)

As highlighted above, current tools for assessing sleep require some level of expertise and/or significant costs. Many companies have leveraged off the popularity of sleep monitoring to create inexpensive and user-friendly monitoring devices that are available to the general public. While CST has some advantages in terms of cost and ease of use, information regarding accuracy and reliability is limited.

3.3.1 Wearable CST

Utilising similar principles to actigraphy, wearable CSTs (e.g. Fitbit, Whoop) allow for the assessment of sleep/wake in the individual's usual environment over long periods of time [17]. These devices are generally relatively inexpensive and some have a range of sensors in addition to accelerometry (oximetry, temperature, light, noise, galvanic skin response) [20]. Unfortunately, the algorithms utilised by these wearable devices are proprietary and very little information is known of the specifics regarding sleep and wake detection [17]. Research comparing some of these commercially available devices to PSG concludes that they tend to overestimate total sleep time and sleep efficiency (surrogate of sleep quality) and underestimate wake after sleep onset [17]. Further, subjects with short sleep times and disturbed sleep demonstrate substantial differences to PSG, and light and deep sleep calculated by consumer CSTs do not correlate with sleep stages measured by PSG [17, 21]. The only published comparison of a CST with polysomnography, in athletes, was conducted by Sargent et al. [21], who compared the Fitbit HR Charge to polysomnography in 12 semi-professional soccer players. When utilising the Fitbit automated scoring algorithm, the average difference in total sleep time between the CST and polysomnography was 51.5 ± 152.4 min. Further, the 95% confidence intervals were extremely wide (−4.2 to 5.9 h). Despite the popularity of newer CSTs such as Whoop, there are currently no published validation studies in any population.

Access to raw data is generally not available for further analysis of CSTs. Ideally, companies would provide access to raw data, which would allow for the application of alternative validated algorithms. These devices, however, are readily accessible and may help educate the individual about aspects of their sleep as well as provide information that may encourage communication and consultation with a professional with sleep expertise. Consumers and practitioners should, however, be aware of the limitations of the devices.

3.3.2 Nearable CST

In a continued bid for compliance and convenience when monitoring sleep, devices are now available that purportedly detect movement when placed on the bed, or near the bed (hence the term 'nearable'). The advantages of these devices are similar to those of wearable CSTs, with the further advantage that the individual does not have to wear the device and therefore may find this type of monitoring more convenient. These devices (e.g. Beddit, ResMed S+) may use ballistocardiography and other forms of limited or non-contact monitoring. As with wearable CSTs, little is known about the algorithms employed with many of these devices and validation against PSG is not comprehensive. Further, device positioning, presence of a bed partner or pet and other extraneous physical factors may impact data quality [20]. Again, the individual should understand the advantages and disadvantages of such technology before ascertaining its usefulness.

3.4 Smartphone Applications

With continually increasing numbers of individuals owning smartphones, consumer-targeted smartphone applications (apps) for sleep self-management are subsequently increasing [22]. There are > 500 sleep-related apps available for download [23]. These apps may use motion detection, audio and video recording as well as questionnaires [24]. Bhat et al. [25] compared the highly popular Sleep Time app with polysomnography and found no correlation between PSG and sleep efficiency ($r = -0.127$; $p = 0.592$), light sleep percentage ($r = 0.024$, $p = 0.921$), deep sleep percentage ($r = 0.181$; $p = 0.444$) or sleep latency ($rs = 0.384$; $p = 0.094$). Sleep Time significantly underestimated light sleep by 27.9% (95% CI 19.4–36.4; $p < 0.0001$), significantly overestimated deep sleep by 11.1% (CI 4.7–17.4; $p = 0.008$) and significantly overestimated sleep latency by 15.6 min (CI 9.7–21.6; $p < 0.0001$).

Apps are generally simple to use, may be free or inexpensive and some provide individualised feedback; this increased awareness has the possibility to induce sustainable sleep enhancements. However, in general, smartphone apps are poor at determining sleep stages and sleep parameters [17], the two steps involved in the monitoring process are combined (data collection and storage) in a 'black box' and researchers have little room to influence or access raw data [26]. Nevertheless, with continual advances in technology, it is possible that Smartphone apps will become an inexpensive and practical option in sleep medicine in the future [26]. For a comprehensive review of current smartphone applications for sleep, see Choi et al. [22].

3.5 Sleep Diaries

The use of a sleep diary can represent a simple and cost-effective means of assessing sleep. Recommended minimum variables to include in the diary are bed and wake time, lights out time, daytime napping, ratings of sleepiness and alertness, caffeine and alcohol intake, exercise and light emitting device use [27] for a duration of at least 1 week. There are also a number of diaries that can be administered via a smartphone application and due to the large variety of sleep diaries available, a consensus sleep diary has been developed by a team of sleep experts [28]. However, biases with diary use may exist due to recall and social desirability and expectation [16] and it is often recommended that diaries are completed in conjunction with activity monitoring.

3.6 Sleep Questionnaires

Sleep questionnaires are another self-assessment tool available to assess a range of sleep variables in athletes. These questionnaires are often utilised as the first diagnostic test due to their ease of administration and low cost. There are a variety of questionnaires available to assess sleep disorders, which may have limited usefulness in athletes. Questionnaires that assess sleep quality (Pittsburgh Sleep Quality Index [PSQI] [29]), assess sleep hygiene (Sleep Hygiene Index [SHI] [30]) or quantify daytime sleepiness (Epworth Sleepiness Scale [31]) may be more useful. Recently, questionnaires designed to specifically examine sleep in athletes have been developed (Athlete Sleep Screening Questionnaire [ASSQ] [32, 33] and Athlete Sleep Behaviour Questionnaire [ASBQ] [34]) and provide useful resources for the elite athlete. For a recent review of available sleep questionnaires, see Ibanez et al. [35].

3.7 Summary

The above information demonstrates that there are potentially a number of advantages and disadvantages of the monitoring tools currently available. An understanding of these pros and cons can aid in the determination of which tool is best suited to the individual's requirements for monitoring and importantly may influence compliance. Table 1 summarises the advantages and disadvantages of monitoring and assessment tools and highlights when their use may be warranted, with consideration given to the following: whether there is suspicion of a medical sleep disorder, research, cost, access to expertise, simplicity of use and validity. It is evident that the market for sleep monitoring is rapidly expanding and it is likely that advances in technology and hardware will continue. As the popular commercially available devices have the potential to greatly affect behaviour, both positively and negatively, the increase in use of these devices should ideally coincide with an increase in research examining the validity and reliability of such devices.

4 Sleep Monitoring Process

There are currently no accepted or standardised processes and procedures for monitoring sleep in athletes. Figure 1 is an attempt to describe a series of steps that may be useful for practitioners who wish to monitor athletes' sleep. It should be noted that this process is best completed in collaboration with a team of professionals, including medical practitioners, psychologists, physiologists, sleep experts and coaches. If medical sleep disorders or mental health concerns are suspected, these should be investigated as a priority.

The process may begin with an initial consultation involving determination of sleep history, including sleep/wake times, sleep environment (noise, light, temperature, presence of bed partner), behaviours (caffeine, training times, phone/computer use) and work/training/life stressors. Asking the athlete to describe their typical 24 h can be useful. The consultation may be combined with or followed by the completion of questionnaires such as the ASBQ, ASSQ, PSQI or SHI. Activity monitoring may be a subsequent step in identifying any sleep concerns and issues regarding routine (consistency of bed and wake times) or is often incorporated during a training cycle as routine monitoring for a 1- to 2-week period of time. Again, if at any time a medical sleep disorder is suspected during this process, referral to a sleep physician should occur. Appropriate feedback, education and behaviour change strategies (see Sects. 6 and 8) should be provided and re-evaluation, follow-up or repetition of the entire process or part thereof may be required.

4.1 Case Study of an Olympic Athlete

The following is an example of the sleep monitoring process conducted on an Olympic athlete (athlete X). Athlete

Table 1 Advantages and disadvantages of sleep monitoring tools

Assessment tool	Advantages	Disadvantages	When to use	Examples
Polysomnography	Gold standard of sleep assessment Allows determination of sleep architecture Identification of medical sleep disorders Useful for assessment of daytime sleepiness	Expensive Does not determine schedules Intrusive unnatural sleep environment Labour intensive Expertise required	Suspected sleep disorder Research	Laboratory or home-based systems
Activity monitoring	Non-intrusive	Does not measure sleep architecture	Research	Philips ActiWatch
Research-grade devices	Less expensive than polysomnography Provides data on routine Can be utilised for extended periods Validated against polysomnography	Does not measure breathing Device can be removed Requires some expertise for analysis More expensive than commercial devices	Monitoring (1–2 weeks)	Fatigue science
Commercial sleep technology	Increase sleep awareness	Lack of validation	Limited expertise and funds available	Fitbit
Wearables	Promote athlete–staff interaction Inexpensive Accessible Can be utilised for extended periods May promote further evaluation	Little information on algorithms utilised Likely to overestimate sleep May cause increased stress and anxiety		Whoop
Commercial sleep technology	Increase sleep awareness	Lack of validation	Limited expertise and funds available	Beddit
Nearables	Promote athlete–staff interaction Inexpensive Accessible May promote further evaluation	Device not worn by individual Apps may increase screen time May cause increased anxiety		Resmed S+ Smartphone applications
Sleep diary	Cost effective Provides information on routine May be more accurate than questionnaires	Requires compliance May be influenced by recall bias May be influenced by subject burden	Limited expertise and funds available Schedule assessment Screening	Multiple available and can be tailored Consensus Sleep Diary
Questionnaires	Cost effective Time effective Can provide information on sleep disorders, daytime sleepiness and sleep hygiene Can provide behaviour information	May be influenced by response bias Lack of standardised data for athletes	Limited expertise available Limited funds available Limited time available Screening	Athlete Sleep Behaviour Questionnaire Athlete Sleep Screening Questionnaire Pittsburgh Sleep Quality Index Epworth Sleepiness Scale Sleep Hygiene Index Visual Analogue Scale Subjective ratings

X mentioned to staff that they were experiencing difficulty in maintaining sleep, perceived their sleep quality to be poor and were waking unrefreshed. The athlete completed the PSQI and recorded a score of 8, indicating poor sleep. Upon analysis of the ASBQ, the athlete did not demonstrate poor sleep behaviours that may have influenced their subjective perception of poor-quality sleep. Based on the outcome of the initial consultation and completion of sleep questionnaires, activity monitoring for a period of 14 days was commenced. The Philips Actiwatch 2 monitor was utilised in conjunction with a sleep diary. The athlete was instructed to wear the monitor at all times with the exception of showering

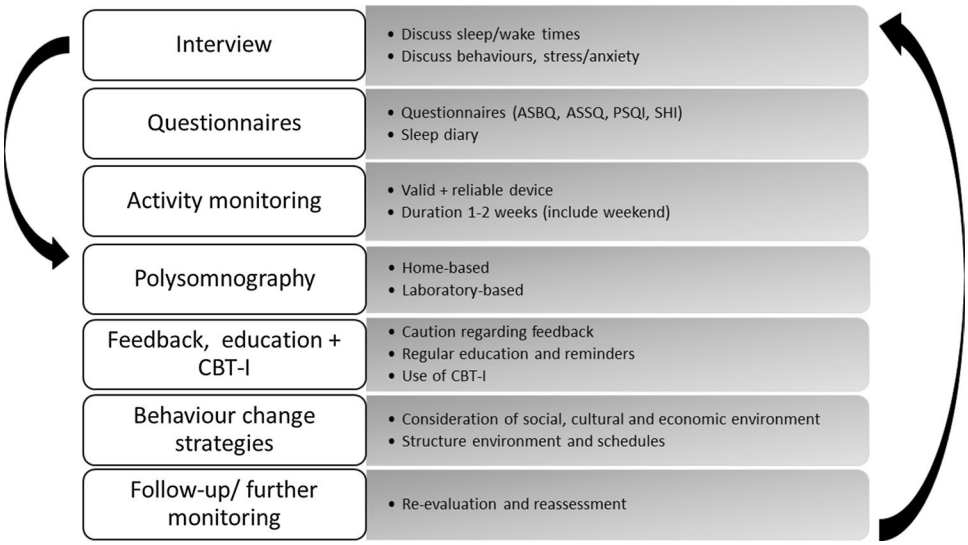


Fig. 1 Suggested process and procedure for monitoring sleep in athletes. *ASBQ* Athlete Sleep Behaviour Questionnaire, *ASSQ* Athlete Sleep Screening Questionnaire, *CBT-I* cognitive behaviour therapy for insomnia, *PSQI* Pittsburgh Sleep Quality Index, *SHI* Sleep Hygiene Index

and swimming, indicate time of attempting sleep and wake (by pressing the button on the activity monitor) and completing the sleep diary for each sleep period (including naps). The results of this monitoring are included in Sect. 6.

5 Precautions Collecting and Interpreting Data

The majority of concerns regarding the collection and interpretation of sleep information are related to CSTs (wearables, nearables and smartphone applications). As mentioned in Sects. 3.3 and 3.4, most either correlate poorly to PSG, have not been compared with the gold standard or, when they have been compared, the devices often overestimate sleep. Further, having ready access to daily information regarding sleep over long periods of time may cause unnecessary stress and anxiety in some individuals. This may actually disturb sleep and exacerbate a poor perception of sleep.

The majority of CSTs are not Food and Drug Administration cleared and are considered lifestyle or entertainment devices/applications. They are often not used for these purposes and may refer to comparisons with PSG. It is important to consider how an unvalidated device serves the consumer. Many individuals assume that data collected objectively are accurate and of more value than subjective data, but this may not be the case with sleep monitoring devices. In addition, are the data able to be interpreted by sleep experts and what are the consequences when an athlete is monitored on separate occasions over time using two different devices that may produce significantly different data outputs? As practitioners we should be aware of the differences between devices and understand how to interpret the data or explain to the individual the reasons not to place excessive importance on aspects of their data. What is clear is that while these devices may facilitate interaction with a professional and encourage healthy sleep behaviours, they should not replace a clinical evaluation or monitoring with a validated technique [22].

In addition, medicolegal concerns, such as publicly available information (cloud storage), courtroom admissibility of data and sharing or tampering with data [36] must be considered. Like other data collected in sport science and sport medicine, issues around ownership and storage must be at the forefront when assessing the elite athlete. Sleep data may include sensitive information and collection and storage of this information in a secure environment is critical.

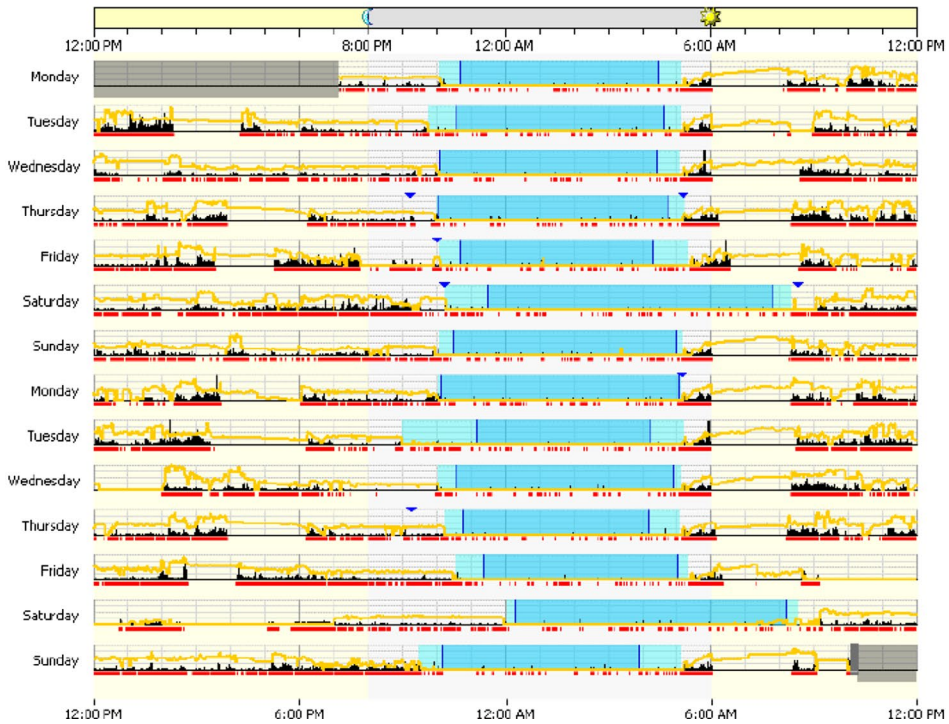
6 Feedback

It is the author's experience that athletes respond best to individual data provided in a simple, easy to read/comprehend format. Due to the relationship between stress, anxiety and sleep, it is important to be cautious when providing feedback regarding poor sleep outcomes as this may perpetuate and reinforce an existing problem. Explaining to the individual that there are specific aspects of their sleep that can be improved and the means to optimise them should be the primary focus, rather than an overemphasis on the suboptimal aspects of the analysis.

Figure 2 incorporates a section of athlete X's sleep report including summary statistics, graphical representation of sleep efficiency, and the 14-day actogram. An explanation of the findings was presented in person to the athlete. Specifically, areas to improve were highlighted to athlete X, including total sleep time, sleep efficiency and time to fall asleep. The interventions prescribed to athlete X are outlined in Sect. 7.5.

Fig. 2 Athlete X sleep report, including summary statistics, graph of sleep efficiency and actogram

	Average	Minimum	Maximum
Bed time	10:08 PM	9:00 PM	12:03 AM
Wake-up time	5:38 AM	5:05 AM	8:32 AM
Total time in bed (h:min)	7:29	6:46	10:05
Time to fall asleep (h:min)	0:37	0:00	2:10
Sleep per night (h:min)	5:10	4:14	7:03
Time awake / light sleep per night (h:min)	1:08	0:50	1:34
Snooze time (h:min)	0:34	0:04	1:12
Sleep efficiency (quality - %)	69	52	77
Your subjective sleep quality rating	Good	Average	Very good



7 Interventions

A range of interventions are available that may assist in enhancing sleep quantity and quality in athletes. Below is a summary of interventions that either may be beneficial or are commonly utilised in athletes. Again, a multidisciplinary approach to treatment and intervention is likely to ensure the best outcome for the athlete.

7.1 Cognitive Behaviour Therapy for Insomnia

The American College of Physicians [4] and the European Sleep Research Society [3] both recommend cognitive behaviour therapy for insomnia (CBT-I) as the first-line treatment for chronic insomnia disorder. This is a consequence of a relatively large body of research on the positive outcomes of CBT-I and a lack of associated adverse effects with this treatment. While many athletes may not present with chronic insomnia, the fundamentals of this treatment may have many benefits for the elite athlete.

CBT-I includes education regarding sleep hygiene, stimulus control therapy, sleep restriction, relaxation techniques and cognitive therapy to reduce rumination about sleep [27]. The sleep hygiene education component includes educating the individual regarding the influence of caffeine, alcohol, light and exercise on sleep as well as information on creating the optimal bedroom environment (minimise light and noise and ensure optimal temperature). Relaxation strategies such as progressive muscle relaxation, meditation and autogenic training may also be included in CBT-I [3, 27]. Behavioural strategies include maintaining a regular routine and getting out of bed if sleep does not occur readily, alongside cognitive strategies such as reducing expectations regarding sleep and reducing unnecessary stress and anxiety around sleep. Challenging and changing misconceptions regarding sleep is important to reduce excessive worry around sleep and the consequences of poor sleep [3].

Computerised versions of CBT-I such as SHUTi and Sleepio are also available, which enhances the practicality of accessing CBT-I [27]. Ideally, when a practitioner is not engaged in face-to face CBT-I, online versions should include personalised feedback to optimise long-term improvements in sleep.

7.2 Sleep Hygiene Education

As mentioned in Sect. 7.1, sleep hygiene is an important component of CBT-I. Sleep hygiene can be described as a variety of different practices that are necessary for normal, quality night-time sleep and full day-time alertness [37] and are outlined in Sect. 7.1. While the effectiveness of CBT-I has not been investigated in athletes, a small number of studies have examined sleep hygiene education in athletes. The results of these studies have indicated increased sleep duration with concurrent increased wake periods [38], and improved subjective sleep duration and efficiency, but no improvements in objective measures of sleep [39].

One study has examined the effectiveness of education and monitoring in professional rugby league players [40]. Following a 2-week period of baseline monitoring, 4 × 30 min education sessions were provided over a 2-week period. The education and monitoring resulted in earlier bedtimes and an increase in sleep duration. However, 1 month following the cessation of the education, sleep behaviours were comparable to those observed at baseline.

7.3 Pharmacotherapy

The most frequently prescribed drugs for insomnia (benzodiazepines and benzodiazepine receptor agonists [BZRA]) have some evidence for short-term (< 4 weeks) treatment of insomnia, but are not recommended for long-term use [3]. Both benzodiazepines and BZRA have potential for tolerance and dependency and there is evidence of significant negative effects on next-day cognitive performance [3]. Further, in certain sports, some medications may not be permitted by anti-doping agencies. Importantly, and surprisingly, there are currently no studies that have addressed the effects of any form of pharmacotherapy on sleep architecture or next-day performance in athletes. The European Sleep Research Society also reports that there is insufficient evidence for recommendations for antihistaminics (strong recommendation, low-quality evidence), antipsychotics (strong recommendation, very low-quality evidence) and melatonin (weak recommendation, low-quality evidence) for the treatment of insomnia [3]. The American College of Physicians report very similar findings to those mentioned above and suggest that physicians use a shared decision-making approach, including discussion of the benefits, harms and costs of short-term medication in adults with chronic insomnia in whom CBT-I alone was unsuccessful [4].

7.4 Other Interventions

Due to the increasing knowledge and awareness regarding potential negative effects of pharmacotherapy for sleep, more 'natural' approaches of sleep interventions have been proposed and minimally investigated. These include light therapy, relaxation, progressive muscle relaxation, neurofeedback/biofeedback and nutrition. While many of these therapies have potential to be effective treatments, there is minimal research currently available. It is likely that this will be an advancing area of research and practitioners should be aware of increasing knowledge in these areas. For reviews

and current evidence base, see Riemann et al. [3], Qaseem et al. [4] and Halson [2].

7.5 Athlete X—Interventions

Initial interventions recommended to athlete X included maintaining a strict bed and wake schedule, minimising caffeine intake prior to less intense training sessions in the afternoon (this may have been causing increased sleep onset latency), lowering bedroom temperature and experimenting with relaxation strategies prior to sleep (to reduce excessive thinking prior to sleep). To determine the effectiveness of these recommendations, another period of follow-up monitoring was scheduled in 4 weeks' time.

8 Behaviour Change

Despite the increasing knowledge and awareness regarding the importance of sleep, many athletes (as well as the general public) continue to have insufficient sleep on a regular basis [41]. Many athletes engage in behaviours that impede night-time sleep (social media, computer games, streaming services [e.g. Netflix], caffeine intake), even in the presence of increased education, monitoring and public awareness of the benefits of sleep for athletic performance.

Only a small number of studies have investigated behaviour change techniques (BCT) as interventions to enhance sleep hygiene [42]. One form of BCT incorporated into sleep research is 'implementation intention', which prompts problem solving and planning and aids in the translation of intention into behaviour [42]. Research has demonstrated that both self-monitoring (using diaries) and implementation intentions appear promising BCTs to improve sleep in undergraduate students [42]. Importantly, very little research exists examining the long-term effectiveness of BCTs for sleep and there is currently no published research conducted in athletes.

The study of Caia et al. [40] mentioned in Sect. 7.2, whereby sleep hygiene education and monitoring improved sleep only for the period of time that the education and monitoring occurred, highlights what is often observed in practice: while initial improvements in sleep may occur in response to education, education in and of itself is not sufficient to result in long-term behaviour change. Therefore, it is important to consider other methods, beyond education and monitoring, to improve sleep behaviours in athletes.

For long-term improvements in sleep behaviours, recent research in behavioural science may provide some insight. Behavioural science is concerned with predicting, explaining and changing behaviour, and in an elite sport environment this may be considered relevant for both athletes and the staff that work alongside them. Behavioural science may enhance our understanding of how to induce change in athletes by examining processes utilised in other health and primary care domains: defining the target behaviour, understanding the psychological determinants of behaviour and developing behaviour change interventions [43].

Importantly, consideration should be given to the fact that behaviour takes place in social environments, which is likely why little progress has been made in decreasing alcohol consumption and preventing dietary- and physical activity-related diseases despite significant public health awareness campaigns [44]. Kelly and Barker [6, 44] describe a variety of common errors when attempting to change behaviour: (i) *Appealing to common sense*—this is ineffective as it ignores the interaction between habit, responses to the environment, conscious choice and culture. Changing behaviour is significantly more complex than common sense. (ii) *Getting the message across is sufficient*—as evidenced by the previously mentioned research in rugby league players [40], education and monitoring does not always lead to long-lasting change. (iii) *Knowledge and information drive behaviour*—information alone, even when provided by experts, does not result in behaviour change. There are usually a number of distractors that prohibit change and acquiring knowledge does not automatically result in a change in the desired behaviour. (iv) *People act both rationally and irrationally*—knowledge and rational assessment of an unhealthy behaviour does not change behaviour as many processes and practices are embedded in daily life and many behaviours serve a purpose in an individual's life. (v) *We can accurately predict behaviour*—individuals may behave very differently in different circumstances and variance in behavioural outcomes is common.

Despite the above-mentioned errors and assumptions, there are a number of techniques that may be useful for changing poor sleep behaviours in athletes. Understanding the environment, culture and economic influences on athletes as well as their underlying motivations can help determine successful behaviour change strategies. Knowledge of patterns and practices occurring before sleep and understanding the behaviours from the individual's perspective (rather than the scientist's) is important. Structuring the environment, including examining schedules (training times, meeting times) can aid in making good choices around sleep by providing optimised sleep opportunities. Minimising distractions and managing stress can also aid in effective behaviour change. At present there is no available literature examining the optimal means of changing sleep behaviour in either athletes or non-athletes and represents an important area of future research.

Importantly, individuals generally do not make decisions in isolation and significant influence lies in relationships with other people, the environment, culture, media, sponsors and sporting organisations. Removing the preconceptions

that sleep is for the weak, or 'I'll sleep when I'm dead' and associating good sleep with optimal health and performance is an important factor. Finally, empowering and motivating the athlete to identify their own solutions to problems will likely result in longer term success.

9 Conclusion

Increasing awareness of the benefits of sleep for athletes has resulted in an increase in sleep monitoring, with little information on standardisation of assessment and feedback. Indeed, we are in the infancy of our understanding of sleep in the elite athlete, which belies the increasing use of both validated and unvalidated technologies. It is essential that practitioners have an understanding of issues that may disturb sleep in athletes, appropriate assessment tools for sleep, advantages and disadvantages of existing and rapidly changing technology and the best means of providing feedback and intervention. Ensuring this process is appreciated and adhered to as closely as possible will help safeguard the athlete from poor quality and potentially unethical monitoring and intervention and may ultimately result in enhanced athletic performance and overall health and wellbeing.

Compliance with Ethical Standards

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